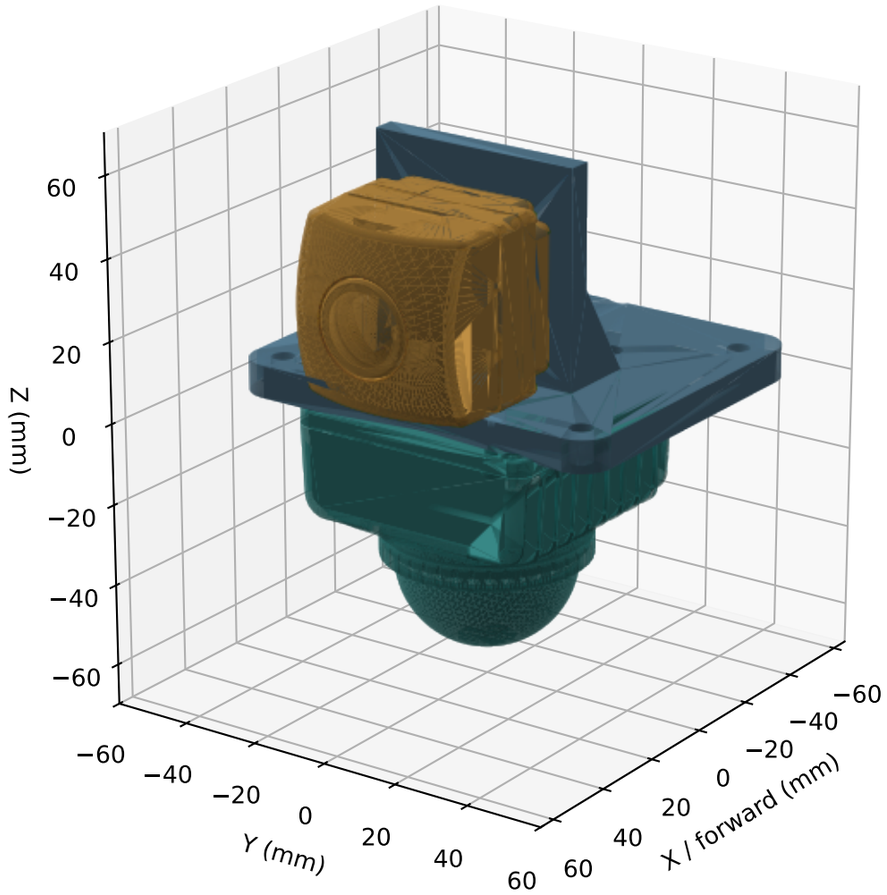
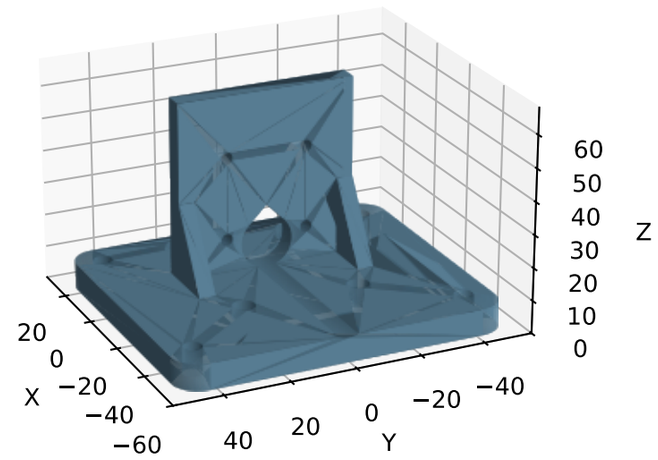
FreeCAD | Rear screw bracket + camera support base

CAMERA ROLL CORRECTED 180 DEG / REAR CONNECTOR LOW / ONE PLA PRINT

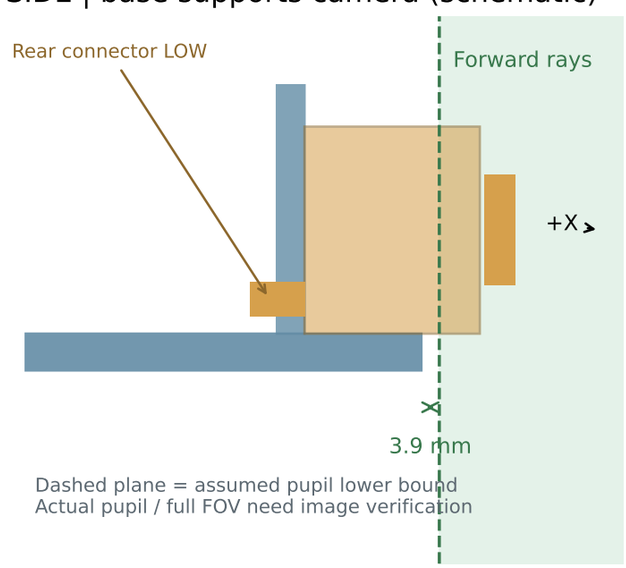
Camera above, inverted MID-360 below



Rear view | 4 screw holes + teardrop cable opening



SIDE | base supports camera (schematic)



Camera underside contact: 612.6 / 668.0 mm2 (91.7%)

Single PLA part + commercial screws. Confirm thread depth, mating plug, actual image and operating temperature.

ONE-PIECE REAR BRACKET | Dimensions and assembly (mm)

FreeCAD native tree and editable Parameters spreadsheet. STL contains fixture only.
Base: 86 x 100 x 8, center (-12,0), R8; front X=31.
Rear plate: 6 thick x 55 wide x 54 above base; Z top=62.
Rear plate camera contact face X=5.6; rear face X=-0.4.
Camera datum X=22 (8 mm rearward from previous X=30), Y=0; bottom Z=8.
Camera roll: 180 deg about forward +X; source +Z points DOWN; rear connector below lens.
MID-360 datum XY=(0,0), mounting face Z=0, scan downward -Z. Both cables exit -X.

CAMERA REAR ATTACHMENT:
 4 x dia 3.6 THRU along X at Y=+/-12.5, Z=18 and 43 (25 x 25 pitch).
 Source STEP rear face Y=-16.4: four entry loops diameter 2.57 at X/Z=+/-12.5.
 M3 thread inferred; axial solid begins 7 mm inside source holes. Verify actual specification.
 M3x10 through 6 mm rear plate gives 4 mm insertion WITHOUT washers (conditional).
 Cable opening Y=0, Z=15.5: dia 16 + 45-degree teardrop roof.
 Base relief below cable: 0.5 deep, extends to rear edge for plug approach.
 Two rear-only gussets: 5 mm wide, 16 mm deep, 26 mm high, with 1 mm root overlaps.

BASE HOLES:
 HOST : (-43,-42), dia 5.7 THRU
 HOST : (23,-42), dia 5.7 THRU
 HOST : (-43,42), dia 5.7 THRU
 HOST : (23,42), dia 5.7 THRU
 LIDAR: (-24,-18), dia 3.6 THRU; TOP counterbore dia 6.8 x 3.5 deep
 LIDAR: (-24,18), dia 3.6 THRU; TOP counterbore dia 6.8 x 3.5 deep
 LIDAR: (24,-18), dia 3.6 THRU; TOP counterbore dia 6.8 x 3.5 deep
 LIDAR: (24,18), dia 3.6 THRU; TOP counterbore dia 6.8 x 3.5 deep

PRINT: PLA; base underside on bed; 0.4 nozzle, 0.2 layers, 5 walls, 6 top/bottom layers.
 40% gyroid and 5 mm brim are starting values. Inspect overhangs/bridges in slicer.
 Start with supports off: cable opening has a 45-degree roof, horizontal screw holes are small.
 Temperature/speed follow filament and printer profiles. Deburr holes and contact faces.

ASSEMBLY:
1. Attach inverted LiDAR to base with 4 screws. M3x8 implies 3.5 mm insertion: verify limit.
2. With rear connector LOW, rest camera on base; slide rearward into cable opening.
3. Fasten 4 rear camera screws from -X. Verify thread, insertion and tightening torque.
4. Connect cables. Attach host via 4 M5 clearance holes; keep LiDAR/cable volumes clear.

CAMERA SUPPORT: 612.6 of 668.0 mm2 bottom planes contact the base.
FOV: fixture Xmax=31; assumed pupil lower-bound X=34.9; rear clearance 3.9.
All fixture material remains behind this plane. Exact pupil and full FOV absent from STEP.
Verify full-frame images, cable plug clearance, PLA temperature/creep, loads and vibration.
Solid PLA mass at 1.24 g/cm3: 105.0 g; actual print includes infill/support effects.